

Strict Positivity on Even Powers Does Not Imply Square Positivity Without Continuity

Candidate counterexample packet
arXiv:1610.01659, Question 2.9

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Status. Candidate full negative answer to the continuity-*removal* part of Question 2.9, for every fixed $d \geq 2$, subject to expert review. The construction does not classify the additional hypotheses weaker than continuity under which the implication might be recovered.

1 Source question

Infusino and Kuhlmann prove that if (A, τ) is a locally multiplicatively convex real algebra and $L : A \rightarrow \mathbb{R}$ is τ -continuous, then the following are equivalent: L is nonnegative on sums of squares; L is nonnegative on sums of $2d$ -th powers for every d ; and it is nonnegative on such powers for at least one d [1, Corollary 2.8]. They then ask on printed page 6:

Does Corollary 2.8 still hold if we remove or weaken the continuity assumption?

$d \geq 2$ (as $M_{2d} \subset M_2$). Thus the positive semidefiniteness of the sequence of Hankel matrices associated to L and M_2 is a sufficient condition for the M_{2d} -positivity of L for any $d \geq 2$. However, we do not know if/when this is also necessary.

For the case $M_{2d} = \sum A^{2d}$ we have the following criterion under the additional assumption of continuity of the considered functional.

Corollary 2.8. *Let (A, τ) be a lmc $\mathbb{R}$ -algebra and $L : A \rightarrow \mathbb{R}$ a τ -continuous linear functional. The following are equivalent:*

- (1) L is positive, i.e. $L(\sum A^2) \subseteq [0, +\infty)$.
- (2) $\forall d \in \mathbb{N}, L(\sum A^{2d}) \subseteq [0, +\infty)$.
- (3) $\exists d \in \mathbb{N}, L(\sum A^{2d}) \subseteq [0, +\infty)$.

Proof. From the discussion immediately after Question 2.7 we know that it is always true that: (1) $\Rightarrow$ (2) $\Rightarrow$ (3). Let us show that also (3) $\Rightarrow$ (1). Suppose that $\exists d \in \mathbb{N}, L(\sum A^{2d}) \subseteq [0, +\infty)$. Then, by Proposition 2.5 applied to $M = \sum A^{2d}$, L has a representing measure μ supported on $X_M \cap \mathfrak{sp}(\tau)$. (In this case $X_M = X(A)$ and so $\text{supp}(\mu) \subseteq \mathfrak{sp}(\tau)$.) In particular, this implies that for all $a \in \sum A^2$ we have $L(a) = \int_{\mathfrak{sp}(\tau)} \hat{a}(\alpha) \mu(d\alpha)$ which is non-negative since μ is non-negative and $\hat{a}(\alpha) \geq 0$ for all $a \in \sum A^2$. Hence, (1) holds. $\square$

Note that while (1) $\Rightarrow$ (2) $\Rightarrow$ (3) holds in general, to show that (3) $\Rightarrow$ (1) we have used the continuity of L . This motivates the following question:

Question 2.9. *Does Corollary 2.8 still hold if we remove or weaken the continuity assumption?*

We show that continuity cannot simply be removed. The counterexample already occurs in the one-variable polynomial algebra.

2 Proof intuition

A functional on $\mathbb{R}[x]$ is determined by its moments $m_j = L(x^j)$. Positivity on $2d$ -th powers says that every finite Hankel form

$$(c_0, \dots, c_N) \mapsto L \left(\left(\sum_{j=0}^N c_j x^j \right)^{2d} \right)$$

is positive semidefinite. This higher-order condition need not make the ordinary Hankel matrix positive semidefinite.

We start in dimension two with a strictly positive binary form whose moment m_2 is negative. To add one polynomial degree, first declare all new moments except the very last one to be zero. The extended form is the old positive-definite form plus finitely many mixed terms plus $M|c_{N+1}|^{2d}$. On the old finite-dimensional coefficient space, the old form has a uniform positive lower bound and each mixed form has a uniform upper bound. Young's inequality absorbs every mixed term into those two pure terms once the new top moment M is sufficiently large. Iteration gives one functional on all polynomials.

3 A positive-definite Hankel-form completion lemma

Fix $d \geq 2$ and write $r := 2d$. For real numbers $m_0, m_1, \dots$, define

$$P_N(c_0, \dots, c_N) := \sum_{i_1, \dots, i_r=0}^N m_{i_1+\dots+i_r} c_{i_1} \cdots c_{i_r}. \quad (1)$$

Lemma 1. *There is a real sequence $(m_j)_{j \geq 0}$ such that every P_N is positive definite, $m_2 = -\binom{r}{2}^{-1} < 0$, and the subsequence $(m_{rN})_N$ is unbounded.*

Proof. For $0 \leq j \leq r$, set

$$m_0 = m_r = 1, \quad m_2 = -\binom{r}{2}^{-1}, \quad m_j = 0 \quad (j \notin \{0, 2, r\}).$$

Then

$$P_1(a, b) = a^r - a^{r-2}b^2 + b^r.$$

Weighted AM–GM gives

$$|a|^{r-2}|b|^2 \leq \frac{d-1}{d}|a|^r + \frac{1}{d}|b|^r,$$

and therefore

$$P_1(a, b) \geq \frac{1}{d}|a|^r + \frac{d-1}{d}|b|^r. \quad (2)$$

Thus P_1 is positive definite.

Suppose moments through m_{rN} have been chosen and P_N is positive definite. Define

$$m_{rN+1} = \dots = m_{r(N+1)-1} = 0, \quad m_{r(N+1)} = M, \quad (3)$$

where $M > 0$ will be fixed below. If $q \in \mathbb{R}[x]$ has degree at most N , the binomial expansion has the form

$$L((q + tx^{N+1})^r) = P_N(q) + \sum_{k=1}^{r-1} t^k A_{k,N}(q) + Mt^r. \quad (4)$$

Here $A_{k,N}$ is already determined before M is chosen: its largest possible moment index is

$$k(N+1) + (r-k)N = rN + k < r(N+1).$$

It is a homogeneous polynomial of degree $r-k$ in the coefficients of q .

Give the space of polynomials of degree at most N any Euclidean coefficient norm. Compactness of its unit sphere and positive definiteness give

$$P_N(q) \geq \delta_N \|q\|^r$$

for some $\delta_N > 0$. Likewise, there are constants $C_{k,N}$ with $|A_{k,N}(q)| \leq C_{k,N} \|q\|^{r-k}$. The weighted Young inequality provides numbers $D_{k,N} < \infty$ such that, for all q and t ,

$$C_{k,N} |t|^k \|q\|^{r-k} \leq \frac{\delta_N}{2(r-1)} \|q\|^r + D_{k,N} |t|^r. \quad (5)$$

Choose

$$M > \sum_{k=1}^{r-1} D_{k,N} + N + 1.$$

Equations (4)–(5) yield

$$P_{N+1}(q + tx^{N+1}) \geq \frac{\delta_N}{2} \|q\|^r + \left(M - \sum_{k=1}^{r-1} D_{k,N} \right) |t|^r > 0$$

unless $q = t = 0$. This completes the induction. The additional $N+1$ in the choice of M makes $(m_{rN})_N$ unbounded. $\square$

4 Main counterexample

Theorem 2. *For every integer $d \geq 2$ there exist a unital commutative normed real algebra (A, ρ) and a linear functional $L : A \rightarrow \mathbb{R}$ such that*

$$L(q^{2d}) > 0 \quad \text{for every } q \neq 0, \quad L(x^2) < 0. \quad (6)$$

Consequently, condition (3) of Corollary 2.8 in [1] does not imply condition (1) after continuity is removed, and L has no nonnegative representing measure.

Proof. Take $A = \mathbb{R}[x]$ with the submultiplicative coefficient norm

$$\rho\left(\sum_j a_j x^j\right) := \sum_j |a_j|.$$

Let (m_j) be supplied by Lemma 1 and define the linear functional L by $L(x^j) = m_j$. Every nonzero polynomial q belongs to a finite degree space, so (1) and positive definiteness of the corresponding P_N give $L(q^{2d}) > 0$. Hence L is nonnegative on $\sum A^{2d}$. On the other hand,

$$L(x^2) = m_2 = -\binom{2d}{2}^{-1} < 0,$$

so L is not nonnegative on $\sum A^2$.

The functional is not ρ -continuous: continuity would imply $|m_j| = |L(x^j)| \leq \|L\| \rho(x^j) = \|L\|$ for all j , contradicting the unbounded subsequence in Lemma 1. Finally, a nonnegative representing measure would give $L(x^2) = \int x^2 d\mu \geq 0$, which is impossible. $\square$

5 Verification, scope, and novelty

The proof was checked at three levels. First, (2) is an explicit coercive bound for the negative-second-moment base case. Second, the degree identity $rN+k < r(N+1)$ guarantees that the coefficient M occurs only in the pure new-coordinate term. Third, the finite-dimensional compactness and Young bounds in (5) make the induction uniform over all old polynomials, not merely pointwise. The accompanying script verifies the base coefficients and this degree bookkeeping for $2 \leq d \leq 10$ and the first thirty extension stages.

This settles the version with continuity deleted, for every $d \geq 2$. It does not decide which particular quasi-analytic, boundedness, or other hypotheses strictly weaker than continuity may restore the implication.

A bounded novelty search through 11 August 2026 covered the run indexes, the exact text of Question 2.9, the source title and authors, later infinite moment-problem papers, and combinations of “positive on $2d$ -th powers,” “square positive,” “Hankel form,” and “continuity.” The search found finite-dimensional positive semidefinite Hankel tensors whose associated Hankel matrices are not positive semidefinite [2, 3], but no global linear functional on $\mathbb{R}[x]$, recursive all-degree completion, or stated answer to Question 2.9. This is a bounded retrieval result, not proof of novelty.

Human-review recommendation. High priority. Check the uniform absorption in (5) and the moment-index calculation in (4). Those are the only nontrivial completion steps; once established, (6) is immediate.

References

- [1] M. Infusino and S. Kuhlmann, *Infinite dimensional moment problem: open questions and applications*, Contemp. Math. **697** (2017), 187–201; arXiv:1610.01659.
- [2] Q. Wang, G. Li, L. Qi, and Y. Xu, *New classes of positive semi-definite Hankel tensors*, Minimax Theory Appl. **2** (2017), 231–248; arXiv:1411.2368.
- [3] W. Ding, L. Qi, and Y. Wei, *Inheritance properties and sum-of-squares decomposition of Hankel tensors: theory and algorithms*, arXiv:1505.02528.